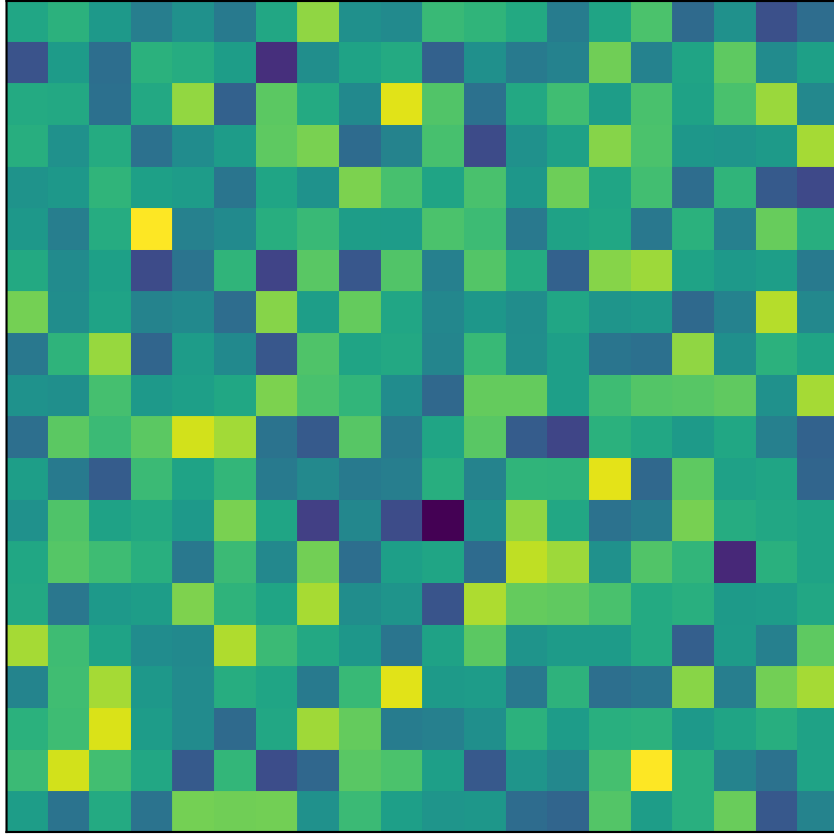
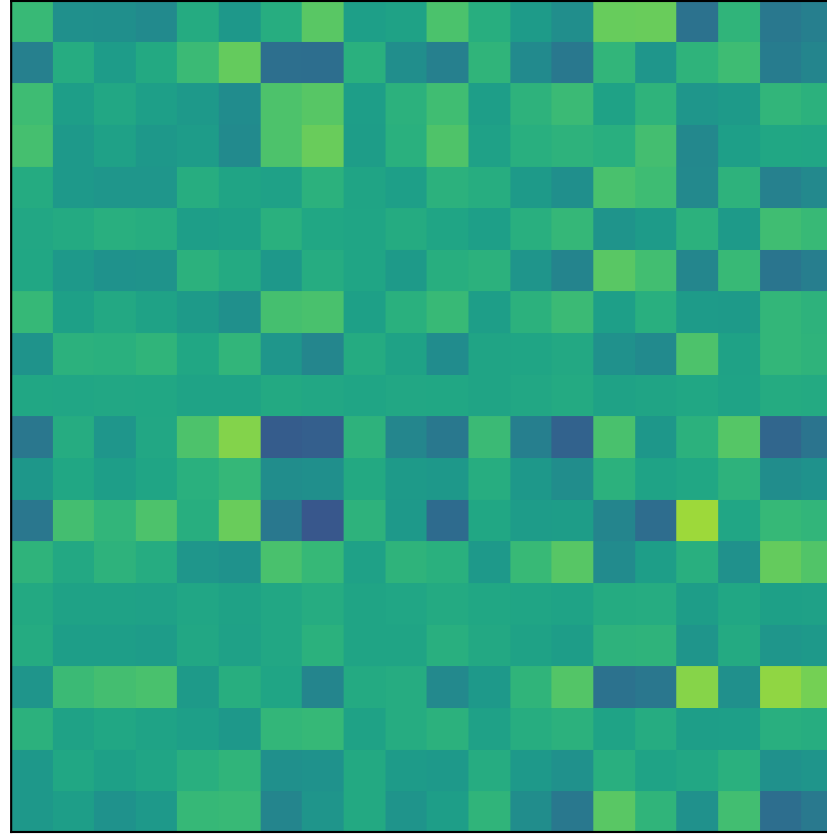


LoRA rank sweep on random full-rank shift
 $M = 20$, LoRA ranks=[1, 2, 4], benchmark=rastrigin

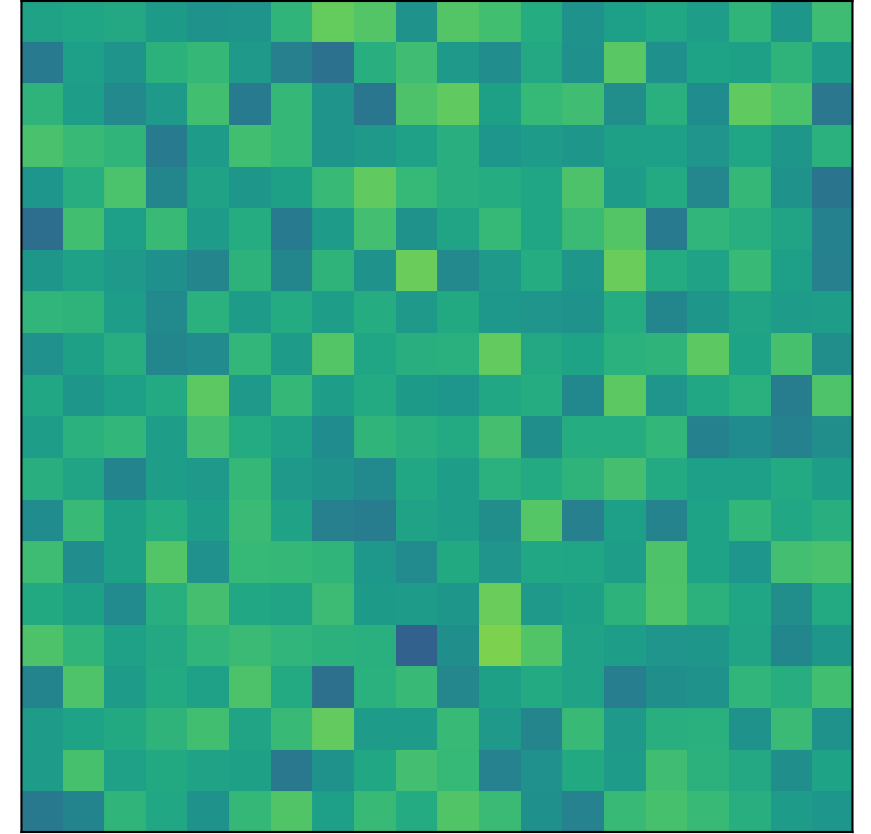
Random shifted optimum X_*



LoRA SVD at $r=2$ \nrel. MSE=7.06e-01



Best RP at $r=2$, $d=80$ \nrel. MSE=7.25e-01



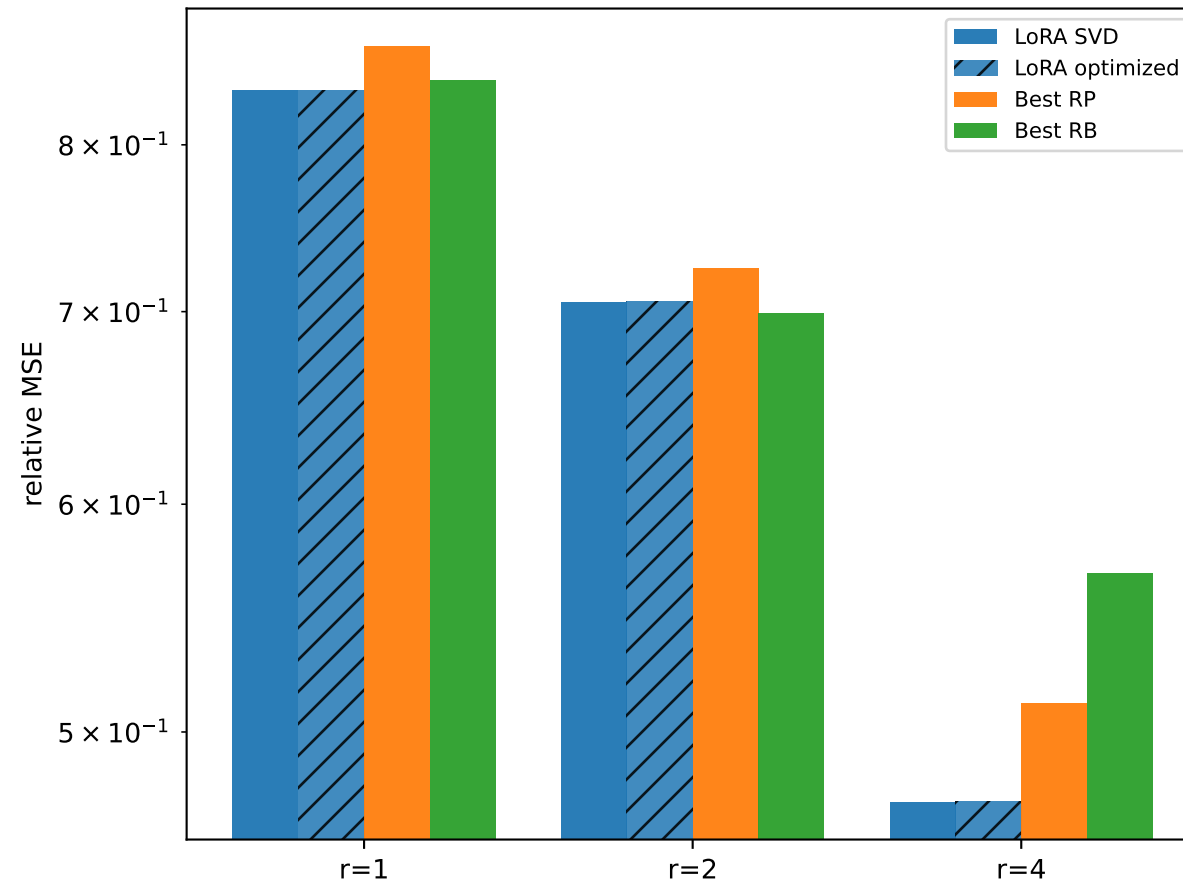
Setup

Random full-rank optimum.

Higher LoRA rank improves the best rank- r approximation (SVD lower bound).

RP/RB baselines are recomputed at each matched search dimension $d = 2Mr$.

Reconstruction error vs LoRA rank



Shifted rastrigin objective vs LoRA rank

